SentinelNet

Decentralised Cybersecurity Threat Detection on Bittensor

Bittensor Subnet Ideathon — Round I Submission

If the exploit fires in the sandbox, intelligence is proven.

> *"SentinelNet turns exploitability into a measurable, cryptographic truth. If the exploit fires in the sandbox, intelligence is proven."*

1. Introduction

SentinelNet is a proposed Bittensor subnet that builds a decentralised cybersecurity threat detection engine. Miners analyse network logs, smart contract bytecode, and system snapshots to identify real security vulnerabilities. Validators confirm every finding by running controlled exploit simulations in isolated sandbox environments — not live systems. The result is an objective, on-chain record of which miners detect real threats, and which do not.

Global cybersecurity spending reached $215 billion in 2024 and is projected to exceed $300 billion by 2027 (Gartner). Despite this, most threat detection is centralised, opaque, and priced out of reach for smaller organisations. DeFi protocols lose billions to exploits that verifiable analysis could have caught. There is no open, cryptographically provable benchmark for security intelligence quality.

1.1 Why This Works as Proof of Intelligence

SentinelNet measures exploitability under reproducible conditions — not prediction accuracy, not opinion, not model confidence. A vulnerability either exists or it does not. A sandbox exploit either fires or it fails. That binary outcome makes scoring fully objective and impossible to fake.

Miners must interpret noisy logs, separate signal from noise, generalise to unseen exploit vectors, and avoid false positives — all under a timed deadline. This is **causal threat inference under adversarial conditions** — the precise kind of structured reasoning that Bittensor's incentive layer is designed to reward.

1.2 Why Bittensor?

SentinelNet is built on Bittensor because no other architecture can deliver these three properties together:

- Decentralised miner competition produces detection diversity. No single vendor controls the model.
- On-chain weights prevent opaque vendor lock-in. Every scoring outcome is auditable.
- TAO emission aligns continuous adversarial improvement. Miners are paid to stay ahead of new threats.

A centralised SaaS, a DAO, or a standard marketplace cannot combine all three. A SaaS is a single point of failure and opacity. A DAO votes too slowly for real-time threat response. A

marketplace cannot enforce objective scoring. Bittensor makes all three possible simultaneously.

2. Incentive and Mechanism Design

2.1 Epoch Lifecycle

Each evaluation cycle — called an epoch — runs every 6 hours. This cadence is not arbitrary. A newly published CVE can be weaponised in under 4 hours. Six-hour epochs create a near-real-time detection loop that is genuinely useful to security teams. They also prevent time-based gaming: a longer cycle would allow miners to reverse-engineer task inputs between submissions. The 6-hour cadence preserves miner incentive alignment with real-world SLA expectations.

The full epoch lifecycle runs as follows:

```
Task Published → Miner Detection → Sandbox Replay → On-Chain Score → TAO Emission
```

Step 1 — Validator Publishes Task	Task type assigned: network log, smart contract bytecode, or system snapshot, with severity threshold defined.
Step 2 — Miners Analyse Input	Miners run detection pipelines against sandboxed data. Findings submitted as structured, signed threat reports.
Step 3 — Submission Deadline Locks	On-chain timestamp checked. Late submissions auto-rejected. Duplicate reports hash-fingerprinted and zeroed.
Step 4 — Validator Runs Controlled Simulation	Findings tested in an isolated Docker sandbox. Exploit confirmed or disproved objectively. No live systems at risk.
Step 5 — False Positive and Negative Check	Was the threat real? Was it missed? Results logged on-chain for full auditability. Ground truth established.
Step 6 — Scoring Computed	F1 Score, Severity Accuracy, Speed — EMA14 applied over rolling 14-day window. Final Score calculated.
Step 7 — Weights and Emissions Updated	On-chain miner weights refreshed. TAO distributed to miners and validators automatically.

2.2 Emission and Reward Logic

Emissions are distributed each epoch based on Final Score rank. The 8% DAO Reserve is allocated to maintaining CVE database feeds, funding sandbox infrastructure, and commissioning independent security audits of the subnet itself.

Recipient	Share	Rationale
Rank 1 Miner	**28%**	Maximum incentive for best detection accuracy.

Rank 2–5 Miners	**28%**	Proportional share — rewards competitive depth across top performers.
Rank 6+ Miners	**14%**	Keeps ecosystem wide, discourages single-miner monopoly.
Validators	**22%**	Compensates the heavy sandbox simulation workload per epoch.
Subnet DAO Reserve	**8%**	Funds CVE database feeds, sandbox infrastructure, and security audits.

2.3 Scoring Formula

Each miner's Final Score uses a clean, three-layer formula:

Final Score = EMA14(Base Score)

Base Score = (F1 Score × 0.60) + (Severity Accuracy × 0.25) + (Speed Score × 0.15)

F1 Score = 2 × Precision × Recall / (Precision + Recall)

Novelty Bonus: +10% emission bonus (applied separately — does not inflate Base Score)

Design principles: Stake is removed from score calculation. It is used only for Sybil resistance and validator admission — not reward weighting. Performance determines reward. Stake determines eligibility. This keeps the formula clean, fair, and resistant to wealth-based gaming.

EMA14 (14-day exponential moving average) smooths short-term volatility. A miner who had a bad day is not destroyed. A miner who cheated once cannot hide it.

Novelty Bonus is awarded as a separate +10% emission bonus for finding a zero-day or previously undocumented vulnerability. This runs parallel to the Base Score rather than inside it, preserving mathematical elegance while still strongly rewarding elite detection.

Worked Example — Full Score Calculation

The table below shows a complete score calculation for Miner X over a 14-day rolling window.

Metric	Value	Calculation	Score
True Positives (14d)	23 of 28 submitted	23 / 28	—
False Positives	3 false alarms	3 / 26 total alerts	—

Precision	TP / (TP + FP)	23 / 26	**0.885**
Recall	TP / (TP + FN)	23 / 25 real threats	**0.920**
F1 Score	Harmonic mean of P and R	2 × 0.885 × 0.92 / 1.805	**0.901**
Severity Accuracy	Correct within 1 severity band	21 of 23 correct	**0.913**
Speed Score	Avg response 4.2 min	1 − (4.2 / 10 max)	**0.580**
Weighted Base Score	F1×60 + Sev×25 + Speed×15	(0.901×0.60)+(0.913×0.25)+(0.580×0.15)	**0.857**
EMA14 Applied	14-day exponential smoothing	EMA smoothing applied	**0.838**
FINAL SCORE			**0.838**

Miner X scores 0.838. Their Speed Score (0.580) is their weakest point. Dropping average response from 4.2 minutes to under 2 minutes would push their Final Score above 0.870, moving them into the top-5 emission bracket.

2.4 Adversarial Defense Matrix

Every attack vector has been modelled. In each case, the cost of cheating exceeds the reward.

Attack Vector	Defense Mechanism	Quantified Effect
False positive spam	Sandbox verification + Precision penalty	3 false positives reduce Precision by ~12%, pushing miner below median — 0% top-bracket emissions.
Missing real threats	Recall metric — missed threats penalised	Missing 3 of 10 real threats drops Recall to 0.70, dropping F1 below 0.780.
Sybil attack	Min 1 TAO stake + copy-hash detection	100 fake miners cost 100 TAO upfront, earn less than 40 TAO total — a 60% guaranteed loss.

Copying another miner	SHA-256 hash fingerprint + 10-min delay check	Duplicate reports receive 0% weight that epoch, every epoch.
Severity inflation	Validator severity cross-check vs CVE database	Error greater than 1 band = severity score of 0 for that finding.
Validator collusion	3 validators minimum, median score used	Outlier more than 15% from median is excluded automatically.
Report replay (old CVEs)	Timestamp + CVE publication date check	CVEs older than 90 days rejected as stale every epoch.
Fabricated report (no real exploit)	Sandbox cannot reproduce = false positive confirmed	Scoring degrades within 3 epochs to below-median — financially self-defeating.

2.5 Economic Security Analysis

The key economic principle is a single clean inequality:

Expected Attack Return < Honest Mining Return

Concrete scenario: Assume the subnet emits 200 TAO per epoch. An attacker runs 50 fake miner identities. Minimum stake cost: 50 TAO. Each fabricated report fails the sandbox and triggers Precision penalties. After 3 false positives, the miner falls below median — earning 0% of top-bracket emissions. Expected attack revenue: approximately 0.1 TAO per epoch per fake miner. Honest miners earn 1.2–3.5 TAO per epoch. The attack earns less than 10% of honest returns and destroys itself within 3 epochs.

Honest mining is the only strategy that produces a positive return.

3. Miner Design

3.1 Miner Tasks and Participation Requirements

Miners receive one of three task types per epoch:

- **Network log analysis** — detecting intrusion patterns and anomalies.
- **Smart contract bytecode analysis** — detecting reentrancy, overflow, access control flaws, and logic errors.
- **System snapshot analysis** — detecting misconfigurations, privilege escalation paths, and exposed services.

Miners must submit in at least 5 of every 7 consecutive epochs to remain in active ranking. Miners who drop below this threshold enter a dormant state and earn zero emissions until they resume.

3.2 Input and Output Format

Each task arrives from the validator as a signed, structured JSON object:

Task Input (from validator):

```
{ "task_id": "epoch_4821_task_003",
  "task_type": "smart_contract",
  "input_hash": "sha256:abc123...",
  "input_data": "<base64 encoded bytecode or log>",
  "deadline": 1734500000 }
```

Each miner's threat report response is structured and cryptographically signed:

Miner Threat Report Output:

```
{ "task_id": "epoch_4821_task_003",
  "miner_hotkey": "5Grwva...",
  "submitted_at": 1734499750,
  "threats": [
    { "type": "reentrancy",
      "location": "contract:0xDEAD..., function: withdraw()",
      "severity": "CRITICAL", "cvss": 9.1,
      "cve_ref": "CVE-2023-29456",
      "action": "Add reentrancy guard before external call at line 142" }
  ], "confidence": 0.91, "signature": "0xabc..." }
```

All submissions are cryptographically signed with the miner's hotkey and timestamped on-chain. The structured threat array allows validators to test each finding independently in the sandbox — a miner who finds 5 threats gets scored on each one separately.

3.3 Performance Dimensions

Dimension	What It Measures	Weight
F1 Score	Harmonic mean of Precision and Recall — single balanced accuracy metric.	**60%**
Severity Accuracy	Correctness of CVSS severity rating within one severity band.	**25%**

Speed Score	Response time from task receipt to submission (10-minute maximum window).	**15%**
Novelty Bonus	Finding a zero-day or undocumented vulnerability earns +10% emission bonus.	**+10% (separate)**

The Novelty Bonus is SentinelNet's most powerful incentive for elite miners. Finding a zero-day earns a 10% emission bonus on top of the standard Final Score weight. This creates a strong incentive for the best miners to push beyond known CVE matching and build genuinely novel detection capabilities.

4. Validator Design

4.1 Sandbox Verification

Validators do not review miner reports — they test them. Each validator runs an isolated Docker container that mirrors the environment the miner analysed. Reported vulnerabilities are replayed inside this container to confirm whether they are real. The container is read-only, network-isolated, and destroyed automatically after each test.

SentinelNet does not test live systems. It tests controlled replicas. Validators only process synthetic data, anonymised public datasets, and explicitly consented audit inputs. All consented inputs are hashed and logged on-chain before use. This boundary — enforced at the protocol level — separates SentinelNet from an attack tool.

No other subnet type uses live exploit replay as its scoring mechanism. That is what makes SentinelNet's Proof of Intelligence structurally unique on Bittensor.

4.2 Validator Onboarding

New validators follow a structured onboarding process. They cannot affect miner rankings until they have demonstrated scoring reliability:

1. Stake Minimum TAO	Reduced 50% for first 60 days to encourage early participation.
2. 7-Epoch Shadow Period	Validator scores are computed and compared — but NOT counted toward miner weights.
3. Accuracy Check	Scores must deviate less than 15% from the existing validator median across all 7 shadow epochs.
4. Full Status Granted	Scores now affect miner weights. Eligible for full 22% emission share.

A minimum of three active validators is required for an epoch to be scored. If the active set drops below three, the epoch is deferred until the minimum is restored. This prevents any single validator from gaining undue influence during a low-participation period.

4.3 Validator Incentives and Slashing

Validators earn 22% of epoch emissions — higher than most subnet designs — to reflect the genuine computational cost of running isolated sandbox simulations for every task. All slashing events are recorded on-chain.

Infraction	Penalty	Recovery
1 epoch downtime	0% reward that epoch	Auto — rejoin next epoch.
3+ consecutive epochs missed	Removed from active set	Re-stake + 7-epoch shadow period.
Score deviation >15% from median	50% weight reduction, 3 days	Auto after 3 days if corrected.
Proven sandbox tampering	Full stake slash + permanent removal	No recovery. Final.

Three validators score each task. The median score is used. An outlier more than 15% from the median is excluded automatically. If all three validators collude, task assignments rotate randomly each epoch — no validator can predict which tasks they will receive. The DAO can also trigger a revalidation audit on any epoch using a randomly selected backup validator set. Disputed slashes — except proven sandbox tampering — can be appealed to a DAO vote within 14 days.

5. Business Logic and Market Rationale

5.1 The Problem

The global cybersecurity market spends over $215 billion per year, yet breaches continue to grow. The average cost of a data breach in 2024 was $4.9 million (IBM). In DeFi alone, over $3.8 billion was lost to smart contract exploits in 2023 (Chainalysis). The problem is not insufficient spending — it is that detection is centralised, opaque, and inaccessible to smaller organisations.

5.2 Competitive Landscape

Existing Tools	vs	SentinelNet Advantage
Crowdstrike / Darktrace — closed, proprietary. No external auditability. $50K–$500K/year.	**vs**	Open + Verifiable. Every finding scored against sandbox result on-chain. Any auditor can reproduce.
Bug bounty platforms (HackerOne) — human-only,	**vs**	Decentralised + Scalable. Global miner competition scales throughput automatically.

slow, inconsistent. No objective scoring at scale.		
LLM code scan tools (Copilot, GPT-4) — generalised, not optimised for threats. No verifiable ground truth.	vs	ML-Native + Objective. Sandbox verification gives deterministic ground truth — not human opinion.
Other Bittensor subnets — none target structured cybersecurity detection at this specificity.	vs	First-mover on Bittensor. No direct competitor subnet. Category-defining position.

5.3 Total Addressable Market

Smart contract auditing is a $1.2 billion annual market growing at 28% per year (MarketsandMarkets, 2024). Enterprise network threat detection is a $35 billion market. Capturing 0.1% of the smart contract audit market alone generates $1.2 million in annual protocol revenue.

5.4 Path to Long-Term Sustainability

SentinelNet has three revenue streams that do not depend on TAO emissions alone:

- **Security-as-a-Service API:** On-demand threat analysis for smart contracts and network logs. Free during beta, paid tiers for production use.
- **Enterprise Audit Marketplace:** Organisations submit their own infrastructure for analysis by top-ranked miners. SentinelNet takes a 10% protocol fee.
- **Bounty Co-Funding:** Protocols post security bounties. The DAO matches with TAO. Real-world bounties attract elite miners, raising overall detection quality.

6. Go-To-Market Strategy

6.1 Phased Rollout

Phase	Timeline	Target Users	Key Incentive	Scope
Phase 1 — Beta	0–3 months	Security researchers, open-source projects	Free API + 1.4× miner multiplier	Smart contracts + common CVE logs
Phase 2 — Growth	3–9 months	Web3 protocols, DeFi auditing firms	Leaderboard + bounty co-funding	Network logs, contract audits
Phase 3 — Scale	9–18 months	Enterprise IT, insurance, regulators	SLA-backed API + licensing	Full enterprise threat surface

6.2 Anchor Community Partners

SentinelNet launches with three named anchor communities:

- **Code4rena** — the largest smart contract audit competition platform. SentinelNet is positioned as a continuous audit alternative between competitions.
- **Immunefi** — bug bounty community with 80,000+ registered security researchers. Free API access during beta, co-branded bounty programme for Phase 2.
- **Public Threat Leaderboard** — a weekly public ranking on X (Twitter) showing which miners detected the most critical vulnerabilities that week, creating organic competitive tension that attracts elite researchers.

6.3 Bootstrapping Mechanics

During the first 90 days, SentinelNet applies a 1.4× emission multiplier for miners who submit in every epoch. Validator stake requirements are reduced by 50% for the first 60 days. A public challenge dataset — a curated library of 500 known vulnerabilities across all three task types — is released two weeks before mainnet launch, allowing miner calibration against ground truth before real emissions begin.

6.4 Managing Beta Signal Quality

Early API users receive lower-quality findings than users who join after competitive miners have optimised. SentinelNet addresses this with a **detection_confidence** field in every API response, calculated from miner track record length, historical false positive rate, and task-type specialisation score. During beta, most findings show confidence scores of 40–65. A clear label explains that confidence reflects model maturity, not certainty. This prevents over-reliance on early outputs and protects user trust through the calibration period.

7. Risk Register

Risk	Likelihood	Impact	Mitigation
Sandbox escape — miner exploits validator environment	**Low**	**High**	Containerised Docker, read-only filesystem, network isolation, automatic teardown after each epoch.
False negative rate too high in early phase	**High**	**Medium**	1.4× bootstrap multiplier + 500-vulnerability public challenge dataset for miner calibration before mainnet.
Regulatory concern: running live exploit simulations	**Medium**	**High**	All simulations in air-gapped environments. Legal opinion obtained before Phase 2. Inputs are synthetic or consented.

Low validator count at launch	**Medium**	**High**	Reduced 50% stake for 60 days + 7-epoch shadow period protects integrity while validator set grows.
Miner collusion on shared threat signatures	**Medium**	**Medium**	Novelty bonus rewards unique findings. Copy detection penalises identical reports 100%.
CVE database staleness	**Low**	**Medium**	CVE feeds pulled daily from NVD. Reports referencing CVEs older than 90 days rejected automatically.

8. Round II MVP Snapshot

SentinelNet is buildable today. The Round II MVP scopes to the minimum set needed to prove the mechanism end-to-end:

Component	Scope
Task type	Smart contract bytecode only (reentrancy, overflow, access control)
Sandbox	Basic Docker orchestration — isolated, read-only, auto-teardown per epoch
Validator set	3-validator minimum with 7-epoch shadow onboarding enforced
Scoring	F1 + Severity Accuracy + Speed — full EMA14 formula active from epoch 1
On-chain weight update	Simple rank-based TAO distribution per epoch via Bittensor metagraph
Launch dataset	500-vulnerability public challenge set released 2 weeks pre-mainnet

Network and smart contract logs are added in Phase 2 once the smart contract pipeline is stable. The full mechanism — sandbox, scoring, slashing, and weight update — runs end-to-end from day one.

Conclusion

SentinelNet turns exploitability into a measurable, cryptographic truth.

Every design decision in this document — the scoring formula, the sandbox, the slashing policy, the bootstrap mechanics — follows from that single principle.

The scoring formula is mathematically clean: F1 score, severity accuracy, and speed — with EMA14 smoothing — are fully defined with a worked example. Stake is removed from scoring entirely, preserving fairness and elegance. The adversarial defense matrix addresses eight specific attack vectors. The economic security analysis demonstrates that gaming the subnet is financially self-defeating within three epochs.

The market is real: $1.2 billion in smart contract auditing alone. The path to revenue does not depend on TAO emissions. Named community partners, a pre-launch challenge dataset, and a working confidence-scoring API complete the go-to-market plan.

SentinelNet turns cybersecurity threat detection into a true Proof of Intelligence — one where the proof is a sandbox exploit that either fires or fails, recorded permanently on-chain for anyone to verify.